



Dr. Wojciech Borek

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Dr. Wojciech Borek, PhD and MSc in Engineering, is an assistant professor in the Department of Engineering Materials and Biomaterials at the Silesian University of Technology in Gliwice, Poland, and a member of the association of Polish Cluster of Innovative Forging Technologies "HEFAJSTOS." His scientific interests include materials science, heat treatment, thermomechanical treatment, plastic deformations, and Gleeble simulations; also, he is a specialist in steels, stainless steel, high-manganese austenitic steels, titanium alloys, light metal alloys. He is an author and coauthor of ca. 100 scientific publications worldwide including 10 chapters in books and more than 30 publications in the Web of Science database; he won 10 awards and honors, national and international; has served, or is currently serving, as a contractor of more than 8 research and didactic projects in Poland and abroad, a reviewer of numerous scientific publications and co-promoter of two doctoral dissertations.

MSc Silesian University of Technology, Faculty of Mechanical Engineering, profile:
Materials science - 13.09.2005

PhD Silesian University of Technology, Faculty of Mechanical Engineering,
profile: Materials Science - 3.11.2010

Scientific field of activity and specialty – high-manganese austenitic steels, light metal alloys, computer science of materials, materials processing technologies, thermomechanical treatment, heat treatment, Gleeble simulations, metal science research methods.

Main research interests: steels, high-manganese austenitic steels, TWIP and TRIP effect, thermomechanical and heat treatment, Gleeble simulations, metal science research methods.

Scientific activity

The total sum publications indexed in Web of Science Database: 29

The total sum of citations: 283

H – Index: 12 (Web of Science)

Profile:

Web of Science: <https://www.webofscience.com/wos/author/record/2506340>

Scopus: <https://www.scopus.com/authid/detail.uri?authorId=25927851500>

Google Scholar: https://scholar.google.com/citations?user=ZqJ_ZSAAAAAJ&hl=pl

Most important papers, including an index of publications in JCR-listed journals and internationally recognised research monographs:

1. L. Sozańska-Jędrasik, W. Borek, J. Mazurkiewicz, Mechanisms of plastic deformation light high-manganese steels of TRIPLEX type, Bulletin of the Polish Academy of Sciences: Technical Sciences, DOI 10.24425/bpasts.2021.137412
2. S.C.A Alfaro, W Borek, B Tomiczek, Welding-Modern Topics Intechopen, 2021
3. B. Krupińska, W. Borek, M. Krupiński, T. Karkoszka, The Influence of Ag on the Microstructure and Properties of Cu-Ni-Si Alloys, Materials 13 (15), 3416, 2020.
4. P. Snopiński, T. Tański, W. Borek, Surface Hardening of AlMg5Si2Mn Alloy through Multi-Axis Compression Using Max Strain, Solid State Phenomena 308, 171-180, 2020,
5. M. Opiela, G. Fojt-Dymara, A. Grajcar, W. Borek, Effect of grain size on the microstructure and strain hardening behavior of solution heat-treated low-C high-Mn steel, Materials 13 (7), 1489, 1, 2020.
6. L. Sozańska-Jędrasik, J. Mazurkiewicz, K. Matus, W. Borek, Structure of Fe-Mn-Al-C Steels after Gleeble Simulations and Hot-Rolling, Materials 13 (3), 739, 2020.
7. J. Dutkiewicz, D. Kalita, W. Maziarz, T. Tański, W. Borek, P. Ostachowski, M. Faryna, Effect of KOB0 extrusion and following cyclic forging on grain refinement of Mg–9Li–2Al–0.5 Sc alloy, Metals and Materials International, 1-11, 5, 2019.
8. J. Dutkiewicz, S. Rusz, O. Hilsner, T. Tański, W. Borek, M. Szlezynger, P. Bobrowski, W. Maziarz, Structure of MgLiAl alloys after various routes of severe plastic deformation studied by TEM, International Journal of Materials Research 110 (1), 24-31, 1, 2019.
9. W. Borek, M. Lis, K. Gołombek, P. Sakiewicz, K. Piotrowski, Effect of plastic deformation rate at room temperature on structure and mechanical properties of high-Mn austenitic Mn-Al-Si 25-3-3 type steel, Archives of Materials Science and Engineering 96 (1), 1, 2019.

10. W. Borek, T. Linek, T. Tański, M. Pancielejko, M. Staszuk, Mechanical and Functional Properties of Cavitation Generators with PVD Functional Coatings Intended for Use in the Cavitation Environment, *Key Engineering Materials* 813, 234-240, 2019.
11. T. Linek, T. Tański, W. Borek, Influence of Applied CrN+ WC/C and WC/C Coatings on the Cavitation Wear Processes of Constructional Elements, *Key Engineering Materials* 813, 62-67, 2019.
12. M. Morawiec, W. Borek, M. Mroziński, A. Grajcar, Combined effects of strain and cooling path on hot deformation response and microstructure of low-carbon structural steel, *Archives of Materials Science and Engineering* 95 (1), 2019.
13. L. Sozańska-Jędrasik, J. Mazurkiewicz, W. Borek, K. Matus, B. Chmiela, M. Zubko, Effect of Nb and Ti micro-additives and thermo-mechanical treatment of high-manganese steels with aluminium and silicon on their microstructure and mechanical properties, *Archives of Metallurgy and Materials* 64, 1, 2019.
14. L. Sozańska-Jędrasik, J. Mazurkiewicz, W. Borek, The influence of the applied type of cooling after eight-stage hot compression test on the structure and mechanical properties of TRIPLEX type steels, *MATEC Web of Conferences* 252, 08005, 1, 2019.
15. T. Tański, W. Borek, M. Król, *Magnesium Alloys: Selected Issue*, IntechOpen, 2018.
16. W. Borek, T. Tański, M. Król, *Cavitation: Selected Issues*, IntechOpen, 2018.
17. J. Dutkiewicz, P. Bobrowski, S. Rusz, O. Hilser, T. Tański, W. Borek, M. Łagoda, P. Ostachowski, P. Pałka, G. Boczek, D. Kuc T. Mikuszewski, Effect of various SPD techniques on structure and superplastic deformation of two phase MgLiAl alloy, *Metals and Materials International* 24 (5), 1077-1089, 7, 2018.
18. S. Ebied, A. Hamada, W. Borek, M. Gepreel, A. Chiba, High-temperature deformation behavior and microstructural characterization of high-Mn bearing titanium-based alloy, *Materials Characterization* 139, 176-185, 2018.
19. L.A. Dobrzański, W. Borek, Thermo-mechanical treatment of Fe-Mn-(Al, Si) TRIP/TWIP steels, *Archives of Civil and mechanical engineering* 12, 299-304, 2012.

Promoter:

1. Co-promoter in two doctoral dissertation, one in progress and planned to be completed in 2021 year, and a second doctoral dissertation completed with public defense on October 14, 2020.
2. Review of one international doctoral dissertations.

Didactic achievements:

Exercises and laboratories of following items: Metallurgy, Fundamentals of materials science, Materials Science with the elements of metals physics, Mathematics, Materials Science, Mechanical properties investigation, Methods of materials investigation, Metallic materials and their heat treatment, Heat and surface treatment, Forging, Thermomechanical treatment, Investigation of the structure and properties of materials; Fundamentals of Computer science, Multimedia Techniques, Computer-assisted selection of materials, BSc Project, MSc Project, supervisor of master's and BSc thesis 16 and 21 semester projects. Co-supervisor of two PhD thesis. Main supervisor of laboratory. Editor and co-editor of 10 books

Most important research projects

- “Organization and development of BSc- and PhD-studies on the area of nanotechnology and material science”, Silesian University of Technology, main contractor, finished,
- „Foresight of surface properties formation leading technologies of engineering materials and biomaterials”, Silesian University of Technology, expert, finished,
- “Structural foundation of counteracting cracking by increase of the cold plastic strain energy margin of the newly developed high-manganese steels of TRIP, TWIP and TRIPLEX types, main contractor, finished.
- “Optimization of grain refinement process up to nano range in Mg-Li alloys of variable crystal structure using severe plastic deformation method” Polish Academy of Sciences Aleksander Krupkowski Institute of Metallurgy and Materials Science, main contractor, end date-August 2018.
- Influence of structural mechanisms and phase transformations to energy margin increase of cold plastic strain and fracture counteraction of newly developed, dynamically or/and meta-dynamically recrystallized steels of TRIP, TWIP and TRIPLEX type.
- “Optimization of grain refinement process up to nano range in Mg-Li alloys of variable crystal structure using severe plastic deformation method” Polish Academy of Sciences Aleksander Krupkowski Institute of Metallurgy and Materials Science, main contractor, finished.

Scientific experience gained in the country and abroad

- Czech Republic, Faculty of Metallurgy and Materials Engineering, VSB – Technical University Ostrava, scientific stay in the frame work of the Visegrad Fund, 2014-2015.
- Germany, Freiberg, 2014, Czech Republic, Ostrava, 2015, Slovak Republik, Lubliana, 2011, Russia, Petersburg, 2012, Ukraine Lviv, Ukraine, Khmielnicky State University 2013, Workshop on the field of material engineering in the framework of the project entitled: „Organization and development of Eng- and PhD-studies on the area of nanotechnology and material science”,
- Italy, Torino, Centro Ricerche FIAT, 200-2006; Ireland, Dublin, Dublin City University, Czech Republic, Ostrava, ERASMUS Staff Mobility - Teaching Assignments by HEI Teaching Staff and by Invited Staff from Enterprises, 2014
- Training: Dynamic thermal-mechanical testing of materials and physical simulation of processes using Gleeble simulator, w Technische Universität Bergakademie Freiberg, Institute of Metal Forming, Freiberg, Germany, 3-5.11.2014

The most important international and national distinctions

- Laureate of a one Lear scholarship of the Visegrad Foundation, Czech Republic, Ostrava, 2014, Slovakia, Zylina 2017, Ukraina 2021,
- Honorary award in the name of Prof. Jan Adamczyk granted by the World Academy of Materials and Manufacturing Engineering AMME for young scientist for outstanding doctoral thesis, 2011.
- First grade Rector Award, „For achievements on the organization field”, Gliwice, 2007, I-st grade Rector Award, „For achievements on the science field”, Gliwice, 2004, 2013,
- Brown medal at the 23rd International Innovation& Technology Exhibition in Kuala Lumpur, 17-19.05.2012,
- Brown medal at the Seoul International Invention Fair 2012 in Seoul, SIIF 2012.
- Silver medal at the iENA International Trade Fair "Ideas-Inventions-New Products, Nuremberg, 2012,
- Brown medal at the 7th International Warsaw Invention Show Invitation,, IWIS 2013”, Warsaw, Poland 2013, and Gold medal on International Warsaw Invention Show Invitation,, IWIS 2015”
- Bronze Medal at the International Invention Fair SIIF 2012, Seul,
- Special award granted by King Saud University, Saudi Arabia 2014,

- Bronze Medal at the INST International Invention Show and Technomart Fair, Taipei, 2012,
- Bronze Medal at the Innova Cities LatinoAmerica International Invention Show 2015, Brazil,
- Gold medal at the Seoul International Invention Fair 2015 in Seoul, SIIF 2015,
- • Diploma from the Minister of Science and Higher Education 2014,
- Gold Medal at the International Exhibition of Technology and Innovation IPITEX 2016, Bangkok, Thailand,
- FIRI DIPLOMA for the BEST INVENTION at the International Exhibition of Technology and Innovation IPITEX 2016, Bangkok, Thailand.